

Lecture 10: Multivariate analysis

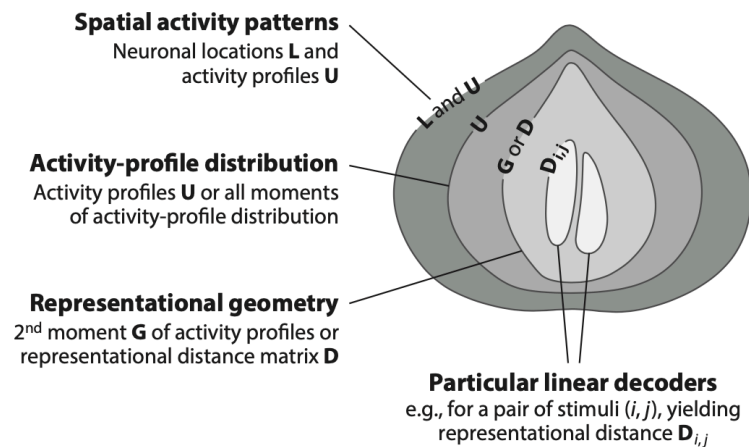
Jochen Braun & Ehsan Kakaei

Advanced Statistics for Neuroscience, SS 2025

1. Neural representations
2. Representational geometry
3. Linear decoding
4. Cross-validation
5. Object recognition
6. Prevalence analysis
7. Multivariate discrimination

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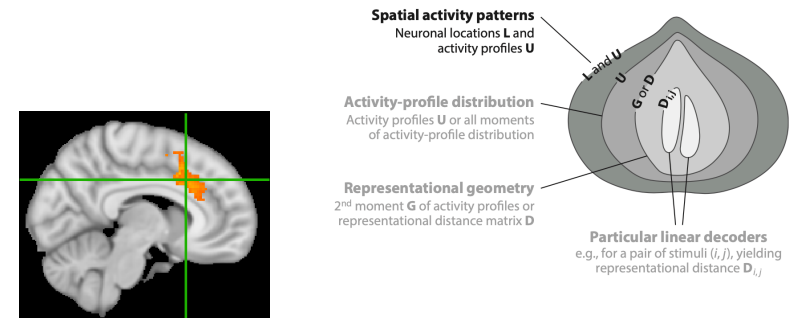
1. The onion of neural representations



Kriegeskorte & Diedrichsen. *Peeling the Onion of Brain Representation*. Annual Review of Neuroscience (2019).

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Spatial activity pattern



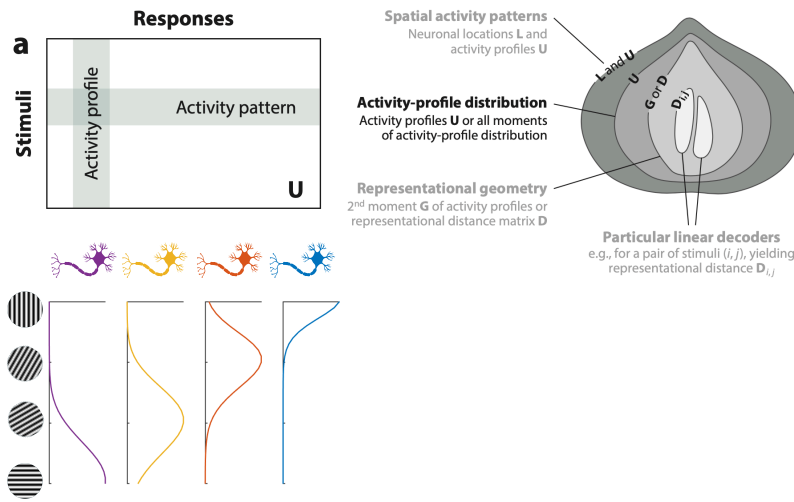
Location matrix $L : [x, y, z]$

Activity profile U

Kriegeskorte & Diedrichsen. *Peeling the Onion of Brain Representation*. Annual Review of Neuroscience (2019).

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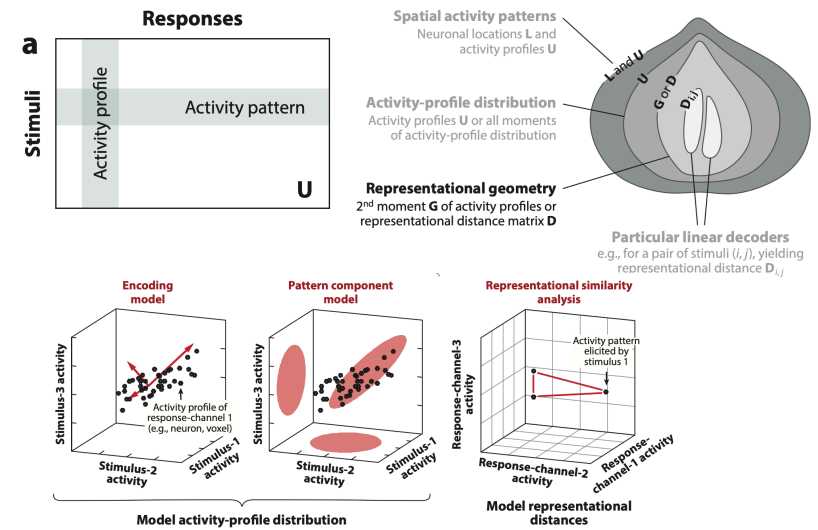
Activity profile and activity pattern



Kriegeskorte & Diedrichsen. *Peeling the Onion of Brain Representation*. Annual Review of Neuroscience (2019).

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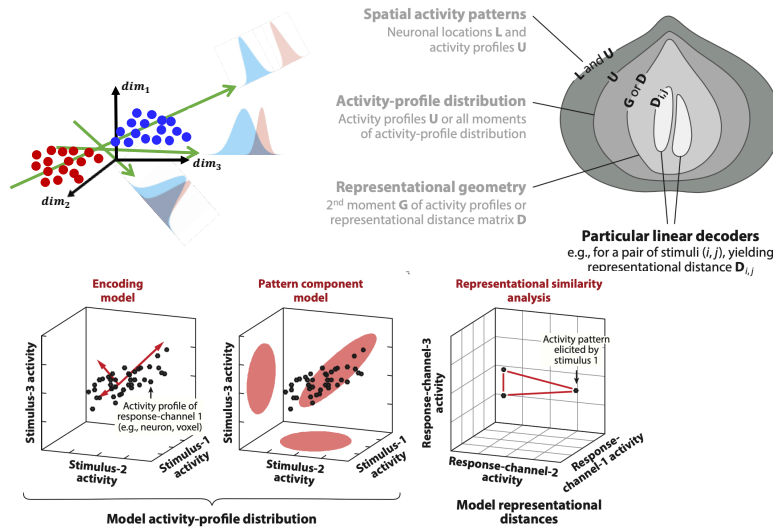
2. Representational geometry



Kriegeskorte & Diedrichsen. *Peeling the Onion of Brain Representation*. Annual Review of Neuroscience (2019).

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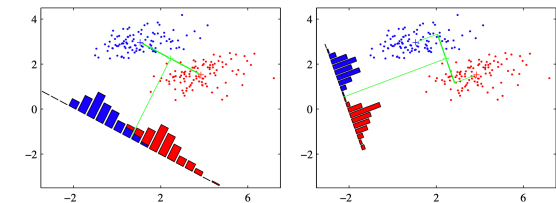
3. Linear decoding



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Fisher's linear discriminant

Linear discriminant analysis



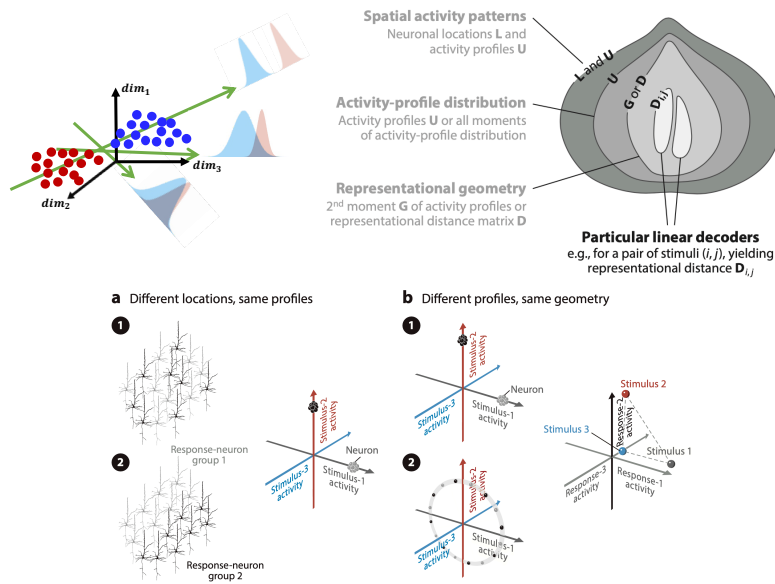
$$C_w = C_1 + C_2 \quad \text{within-class variance}$$

$$y = C_w^{-1}(m_1 - m_2) = (m_1 - m_2)C_w^{-1}$$

$$w = \frac{y}{||y||} \quad \text{Fisher's linear discriminant (FLD)}$$

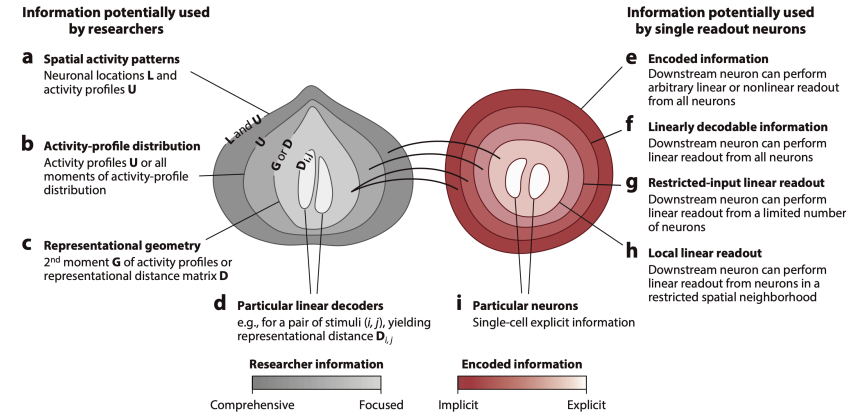
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Equivalent encoding



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Levels of decoding



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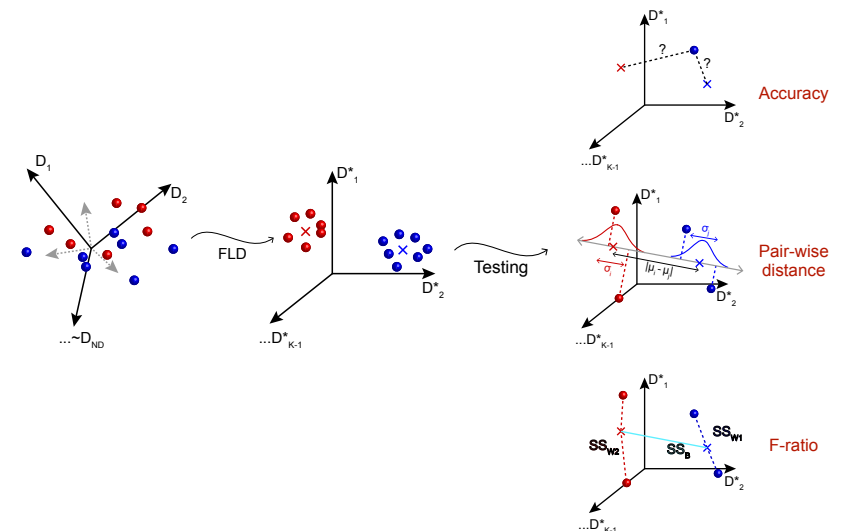
4. Cross-validation required!

- If number of dimensions N is higher than number of classes K , discriminants would overfit the weights.
- Even non-existing classes would be discriminated!
- Cross-validate the discriminant by keeping a portion of the dataset as test set.



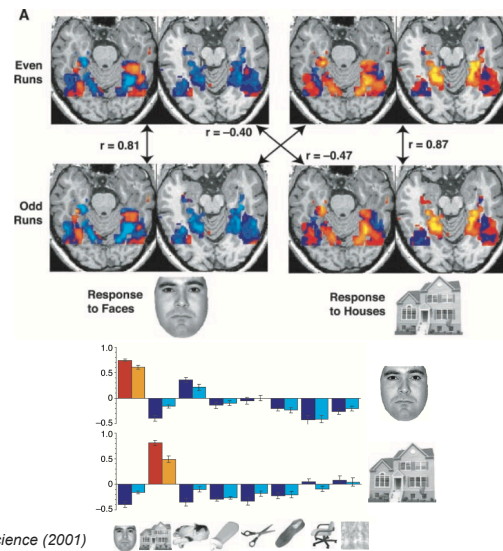
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Cross-validation assessment



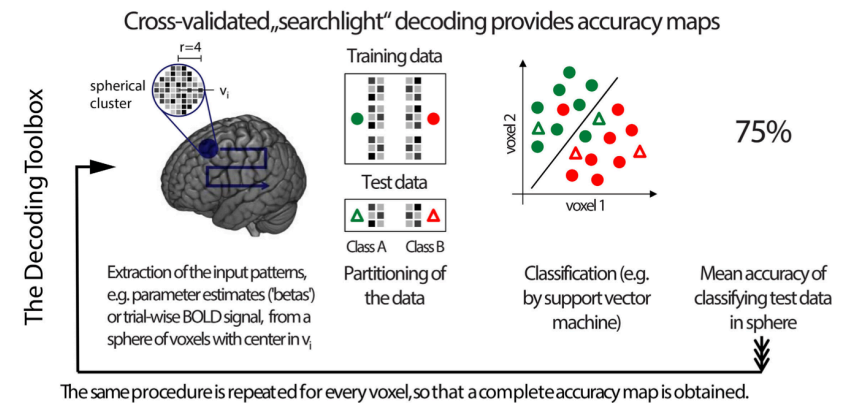
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5. Object recognition results



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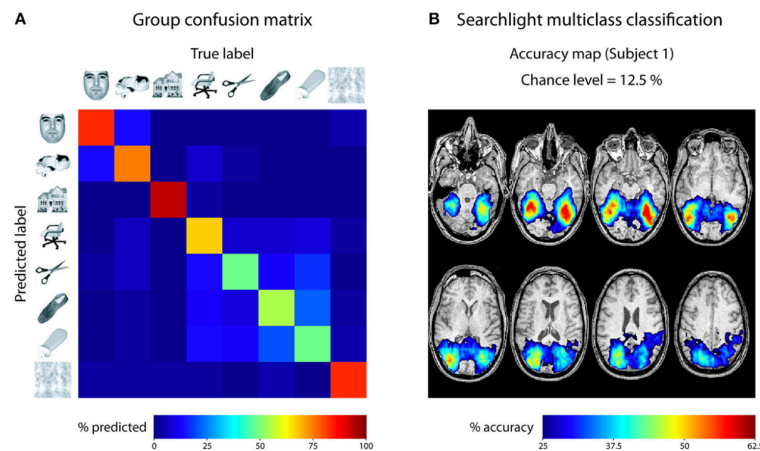
Example: Searchlight decoding



Hebart et al. Frontiers in neuroinformatics (2015)

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Example: Object recognition with searchlight



Hebart et al. Frontiers in neuroinformatics (2015)

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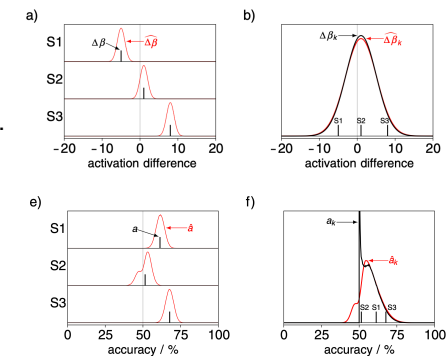
6. Information-based prevalence inference

- Subject and population-level analysis:

1. Subject-level activation
2. Population inference (subjects as random effect), usually *t*-test or ANOVA

- Information cannot be negative! True accuracy is never lower than chance!

X *t*-test



Allefeld et al. (2016) Valid population inference for information-based imaging. *NeuroImage* 141: 378-392.

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Prevalence inference by minimum statistics

- Global null hypothesis: There is no effect in any subject! For all subjects, accuracy is at chance.
- “Prevalence”: in what fraction of the population there is an effect?

Step 1: Test for the global null hypothesis

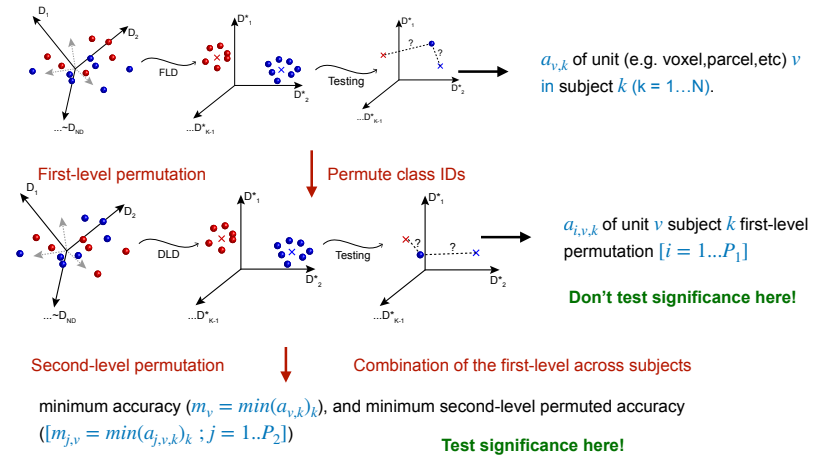
Step 2: Test for the null hypothesis that prevalence is smaller than a threshold.

Allefeld et.al. NeuroImage (2016)

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Prevalence inference

Step 1: Test for the global null hypothesis



Allefeld et.al. NeuroImage (2016)

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Prevalence inference

↓ Unit-level p-value

Uncorrected p-value

$$P_v = \frac{1}{P_2} \sum_{j=1}^{P_2} [m_v \leq m_{j,v}]$$

Corrected p-value over multiple subjects

$$P_v^* = \frac{1}{P_2} \sum_{j=1}^{P_2} [m_v \leq M_j] ; M_j = \max(m_{j,v})_v$$

Step 2: Test for the null hypothesis that prevalence is smaller than a threshold.

$$\text{corrected significance threshold } \alpha_v^* = \frac{(\alpha - P_v^*)}{(1 - P_v^*)}$$

$$\text{prevalence } \gamma_v = \frac{\sqrt{\alpha_v^*} - \sqrt{P_v}}{1 - \sqrt{P_v}}$$

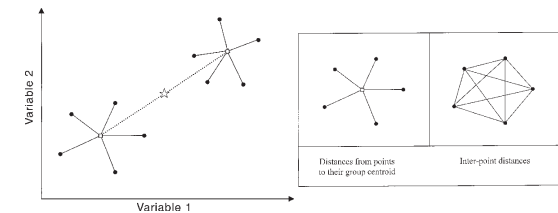
There is an effect if $\gamma_v > \gamma_0$

Allefeld et.al. NeuroImage (2016)

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6. Multiclass discriminant analysis

Multidimensional analysis of variance



$$SS_B = \sum_{k=1}^K n_k (c_k - c_{overall})^2$$

$$SS_W = \sum_{k=1}^K \sum_{i=1}^{n_k} (x_{ik} - c_k)^2$$

$$F = \frac{SS_B / (K - 1)}{SS_W / (N - K)} \quad \text{F-ratio}$$

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7. Multiclass discriminant analysis

Fisher's optimization criterion:

The aim of FLD is to maximize the ratio J of the between-class scatter (SS_B) to within-class scatter (SS_W).

$$J = \frac{w^T SS_B w}{w^T SS_W w}$$

$$\frac{\partial J}{\partial w} = 0 \rightarrow SS_B w = \lambda SS_W w \quad \text{Generalized eigenvalue problem}$$

solution in matlab: $[w, \lambda] = \text{eig}(SS_B, SS_W)$

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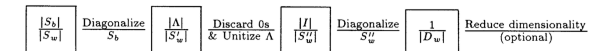
Fisher's linear discriminant in high-dimensional data

- For high-dimensional data FLD is both computationally expensive and scatter matrices are singular!
- Solution is to **reduce the dimensionality of data**.

X PCA+FLD

Direct LDA

! PCA can discard discriminative information

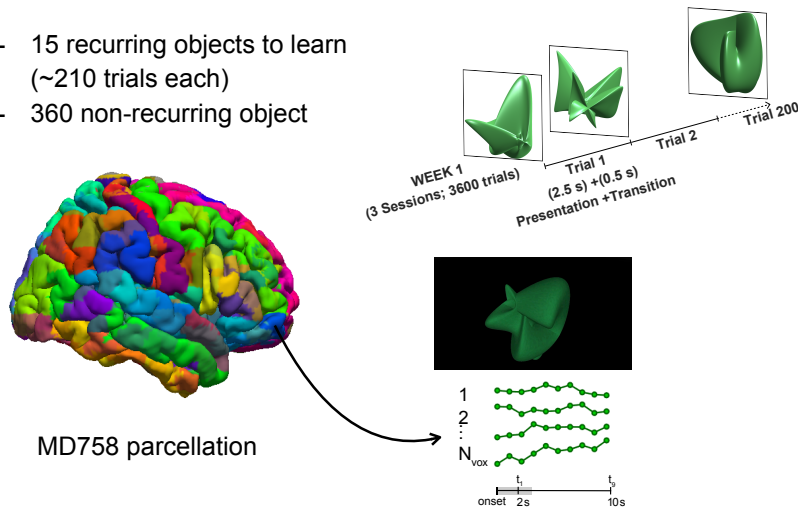


Yu & Yang. Pattern Recognition (2001)

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Example: Object identity representation

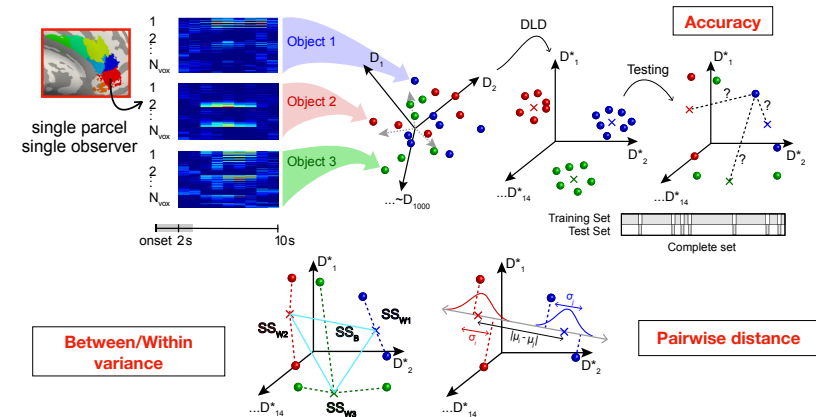
- 15 recurring objects to learn (~210 trials each)
- 360 non-recurring object



Kakaei & Braun (2024) Gradual change of cortical representations ... *Imaging Neuroscience*, Vol. 2
https://doi.org/10.1162/imag_a_00255

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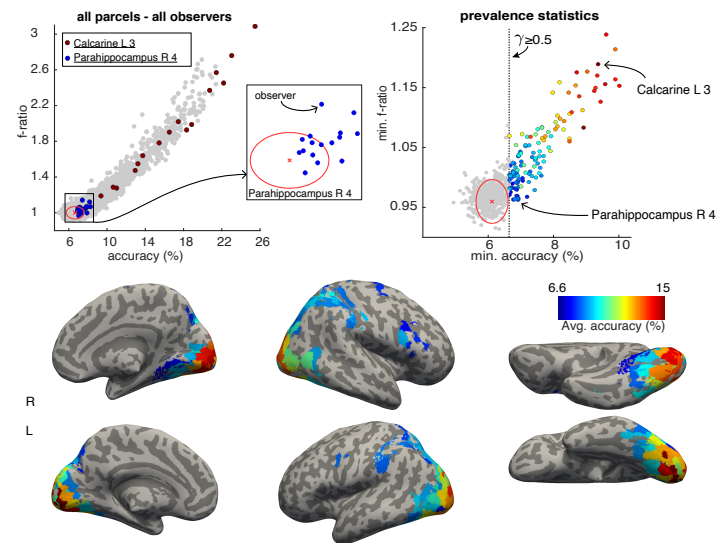
Example: Representational similarity analysis



Kakaei & Braun (2024) Gradual change of cortical representations ... *Imaging Neuroscience*, Vol. 2
https://doi.org/10.1162/imag_a_00255

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Example: Prevalence statistics



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The End